

Jinghan Yang

Ph.D. in Computer Science and Engineering

✉ jinghanyang2023@gmail.com
🌐 <https://jinghany.github.io/website>
in Jinghan Yang
🔗 [jinghanY](#)

Interests

My research focuses on both the performance and robustness of computer vision systems, with an emphasis on autonomous driving. Recently, I have concentrated on designing techniques to improve the effectiveness of video panoptic segmentation models on large-scale autonomous driving datasets. Throughout my previous research, I studied the vulnerabilities and limitations of deep learning models in vision and autonomous driving, with a focus on developing methods that remain reliable under real-world conditions.

Experience and Education

2023–present **Independent Researcher.**

Focused on robustness in computer vision systems as well as panoptic video segmentation.

2017–2023 **Ph.D. in Computer Science and Engineering** Washington University in St. Louis (St. Louis).

Thesis: Physically Realizable Attacks and Defenses for Autonomous Driving Systems.

Advisor: Yevgeniy Vorobeychik

2014–2016 **M.S in Applied and Financial Mathematics** Northeastern University (Boston).

Advisors: Javed A. Aslam and Virgil Pavlu.

2010–2014 **B.S in Financial Mathematics** Tianjin University of Finance and Economics (Tianjin).

Advisors: Wei Geng and Shuhua Zhang.

Research and Projects

2025–present **#1 on the Waymo Open Dataset (WOD) 2D Video Panoptic Segmentation Leaderboard (as of Sep. 2026).**

- Ranked #1 of all leaderboard submissions on all three metrics (wSTQ & wAQ & mIoU). This method is designed for large scale video panoptic segmentation. The competition focuses specifically on multi-camera data from WOD's Perception Dataset: training sequences have 5 labeled temporal frames, and evaluation sequences have about 100 temporal frames sampled at 5 Hz, across 5 cameras (3 front cameras at 1920×1280 , 2 side cameras at 1920×886) with 28 semantic classes.
- My method is built on the DVIS++ offline model with two contributions: (1) a training scheme with a special "hard-region" loss that dynamically corrects model errors on difficult-to-predict pixels; and (2) a decoupled test-time inference pipeline that recovers pixels the tracker drops.

2023–2024 **Robust Environmental Design For Computer Vision:** [Independent Research] This project tackled robustness of computer-vision from an under-explored direction; rather than training a model to be robust, the environment can be optimized to decrease the effectiveness of attacks. I developed a technique for environmental optimization which can take an attacker's success rate from 100% on the unmodified environment to 0% on the modified environment [2].

- 2020-2023 **Robustness for Self-Driving:** [Ph.D. Research] This research consisted of multiple projects aimed making self-driving models more robust by identifying and addressing their weaknesses. Projects in this direction include a method for fast and differential attacks in dynamic online environments (e.g., inside autonomous driving simulators) [5], a method for certified robust control in end-to-end driving systems [4], as well as a method for adversarially controlling large fleets of autonomous vehicles [3].
advisor: Yevgeniy Vorobeychik
- 2019-2019 **CARLA Autonomous Driving Challenge:** [Ph.D. Research] In this project I developed a self-driving model for the CARLA simulator. My model won 3rd place at CVPR Autonomous Driving Challenge .
advisor: Yevgeniy Vorobeychik
- 2018-2022 **Robustness in Computer Vision Systems:** [Ph.D. Research] My research consisted of multiple projects focused on fixing vulnerability of computer vision systems. In particular I developed techniques for protecting geo-location privacy [7] and preventing deep-fakes [6] .
advisor: Yevgeniy Vorobeychik
- 2016-2017 **Automated Hyperparameter Selection in NLP:** [Masters Research] This project focused on developing a pipeline for efficiently selecting hyperparameter used in learning pipelines for natural language classification.
advisors: Javed A. Aslam and Virgil Pavlu
- 2013-2014 **Fluctuations and Market Pricing:** [Undergraduate Research] This project used large scale data from international markets to model and analyze impacts of trading on fluctuations in market pricing .
advisor: Wei Geng
- 2012-2013 **Macroeconomic Analysis:** [Undergraduate Research] This project focused on capturing the correlations between elements of macroeconomics and measurements of climate change; we used these correlations to simulate future behavior using Matlab .
adivsor: Shuhua Zhang

Publications

[1] Hard-Region Supervision: #1 on the Waymo Open Dataset 2D Video Panoptic Segmentation Leaderboard (Technical Report, arXiv 26).

Jinghan Yang

<https://arxiv.org/abs/2609.38714>

[2] RED: Robust Environmental Design (NeurIPS SATA 24).

Jinghan Yang

<https://arxiv.org/abs/2411.17026>

[3] Location Spoofing Attacks on Autonomous Fleets (VehicleSec 23).

Jinghan Yang, Andrew Estornell, Yevgeniy Vorobeychik

[4] Certified Robust Control under Adversarial Perturbations [IEEE ACC 23].

Jinghan Yang, Hunmin Kim, Wenbin Wan, Naira Hovakimyan, Yevgeniy Vorobeychik

[5] Finding Adversarial Examples for Simulated Autonomous Driving with Fast and Differentiable Image Compositing (AAAI TRASE 22).

Jinghan Yang, Adith Bloor, Ayan Chakrabarti, Xuan Zhang, Yevgeniy Vorobeychik

[6] PROVES: Establishing Image Provenance using Semantic Signatures (WACV 21).

Mingyang Xie, Manav Kulshrestha, Shaojie Wang, **Jinghan Yang**, Ayan Chakrabarti, Ning Zhang, Yevgeniy Vorobeychik

[7] Protecting Geolocation Privacy of Photo Collections (AAAI 20).

Jinghan Yang, Ayan Chakrabarti, Yevgeniy Vorobeychik

Contributed Talks

Oral Presentation for "Certified Robust Control under Adversarial Perturbation" at IEEE-ACC 23.

Oral Presentation for "A Fast and Differentiable Adversarial Testing Framework for Simulated Autonomous Driving" at AAAI 22.

Oral Presentation for "Protecting Geolocation Privacy of Photo Collections" at AAAI 20.

Oral Presentation for our developed autonomous driving vehicle in "CARLA Autonomous Driving Challenge" at CVPR 19.

Programing and Software

Open-Source Software Projects

- 2023 **Certified Robust Control under Adversarial Perturbations** This software provides a self-driving agent (comprised of both a perception and control module) which is certifiably robust to adversarial noise. This agent can be deployed in any vision-based autonomous driving simulator and physical world. This software accompanies our IEEE-ACC paper of the same name .
<https://github.com/jinghanY/RobustControl>
- 2021 **Finding Adversarial Examples for Simulated Autonomous Driving with Fast and Differentiable Image Compositing** This software provides a differentiable proxy for the CARLA autonomous driving simulator. This allows researchers to more efficiently study attacks and defenses for simulated autonomous driving by subverting the computational issues which arise from the non-differentiability of simulators. This software accompanies our AAAI TRASE paper of the same name .
<https://github.com/jinghanY/physicalAttackImageComposition>
- 2020 **PROVES: Establishing Image Provenance using Semantic Signature** This software provides users with a learning framework which can certify facial images in order to prevent their use in constructing deep-fakes This software accompanies our WACV paper of the same name .
<https://github.com/jinghanY/imgprov>
- 2020 **Protecting Geolocation Privacy of Photo Collection** This software allows users to protect the geolocation of their publicly shared photo collections. Using deep learning models we modify the photo collection such the geolocation is no longer identifiable. This software accompanies our AAAI paper of the same name .
<https://github.com/jinghanY/geoPrivacyAlbum>
- 2019 **Autonomous Driving Agent for the CARAL Simulator** This software provides a self-driving agent for use in the CARAL simulator (achieved third-place in CVPR CARLA Autonomous Driving Challenge 2019).
<https://github.com/jinghanY/AutonomousDrivingAgent>

Honors and Awards

- 2019 3rd-place in the CARLA Autonomous Driving Challenge.
- 2013 National Scholarship for Outstanding Undergraduates. (only 1 student per year)
- 2013 Tianjin University in Finance and Economics Scholarship for the Rank-1 student.
- 2012 Tianjin University in Finance and Economics Scholarship for the Rank-1 student.
- 2012 1st Place in China's National Undergraduate Mathematical Modeling Contest.

Internships

Jan.-Apr. **Quantitative Analyst Intern.**

2014 Sun and Bright Asset Management Company, Beijing, China

I was responsible for developing statistical tools, based on modified Black-Scholes Equations, which analyzed the financial reports of companies in order to determine optimal strategies

July **Data Analyst Intern.**

2012 MassMutual Financial Group, Hong Kong, China

I was responsible for developing statistical tools which were used to analyze user questionnaires regarding insurance coverage and pricing.

Academic Development

Teaching

2021 Teaching Assistant for *Special Topics; Responsible AI*. (Washington University in St. Louis)

Peer Reviews

CPRV, WACV, ICLR, ICML, NeurIPS, AAAI, KDD , AAMAS , ICDM

Programming Languages/Libraries

Fluent: Python, Tensorflow, Pytorch